LLM PROGRAMMING TASK 20 · HEADLINE GENERATION

Grounded Headline Generation from Public Event Images

Headlines that say only what the photo shows.

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PROBLEM AND OBJECTIVES

Models invent facts that a photo cannot support

News desks need quick headlines for large photo collections.

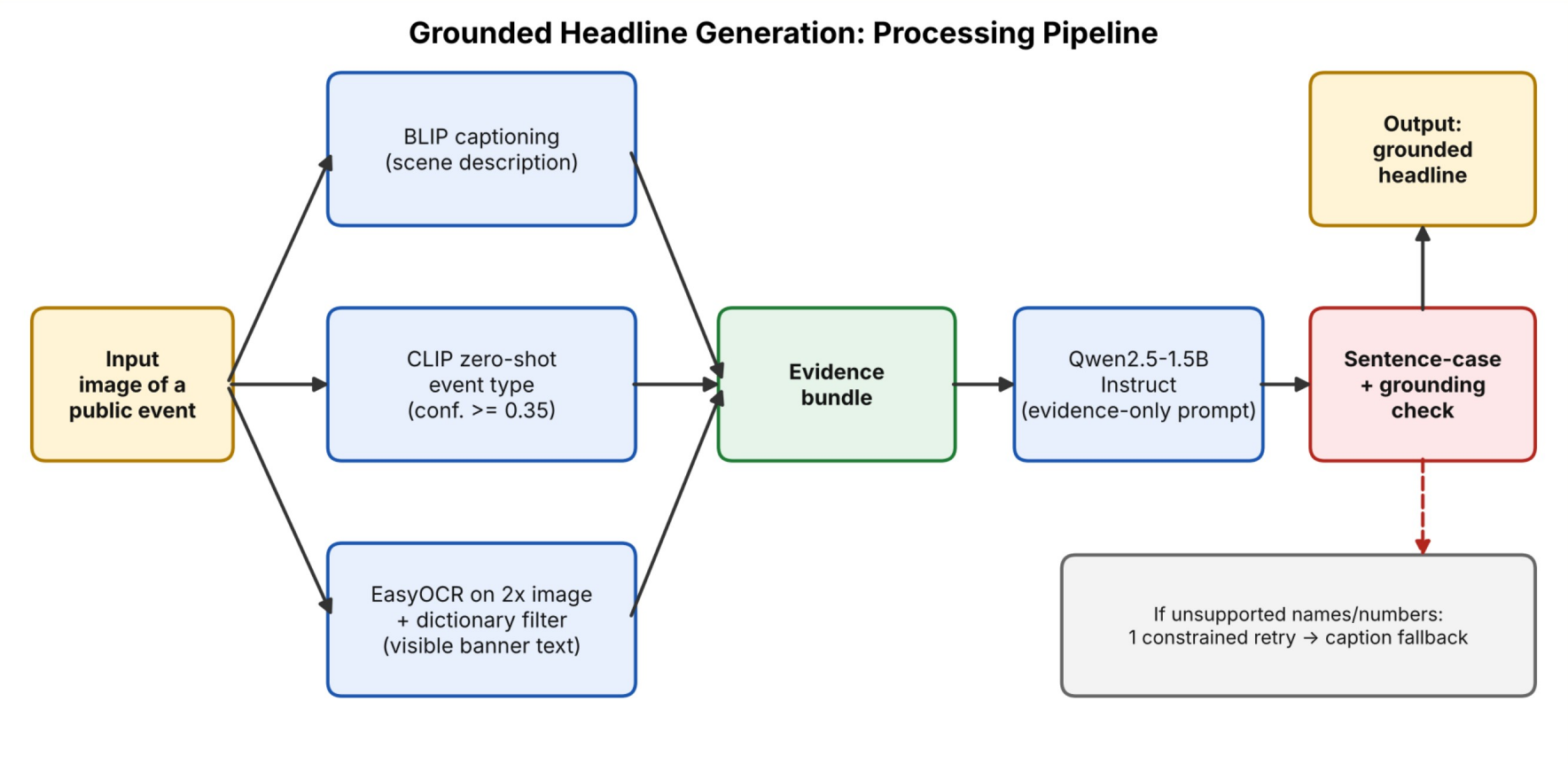
Language models add names, places, dates and causes that are not in the picture. In journalism, that is unacceptable.

Our objectives

- Write a short, fluent headline from one image
- Use only what is visible: the scene, the activity and legible text
- Detect unsupported names and numbers automatically

PROCESSING PIPELINE

Three evidence sources feed one careful writer



STEP BY STEP

How a headline is made

1 · CAPTION

Describe the scene

BLIP writes one sentence about what the photo shows.

2 · EVENT TYPE

Guess the event

CLIP scores 15 generic labels. Weak guesses, under 0.35, are ignored.

3 · VISIBLE TEXT

Read banners

EasyOCR on a 2x image, keeping only real dictionary words.

4 · WRITE

Headline from evidence

Qwen2.5 may use only the evidence in its prompt.

5 · FORMAT

Sentence case

Fixes ALL CAPS and Title Case output.

6 · CHECK

Grounding check

Every name or number must be in the evidence, else retry, then caption fallback.

MODELS AND TOOLS

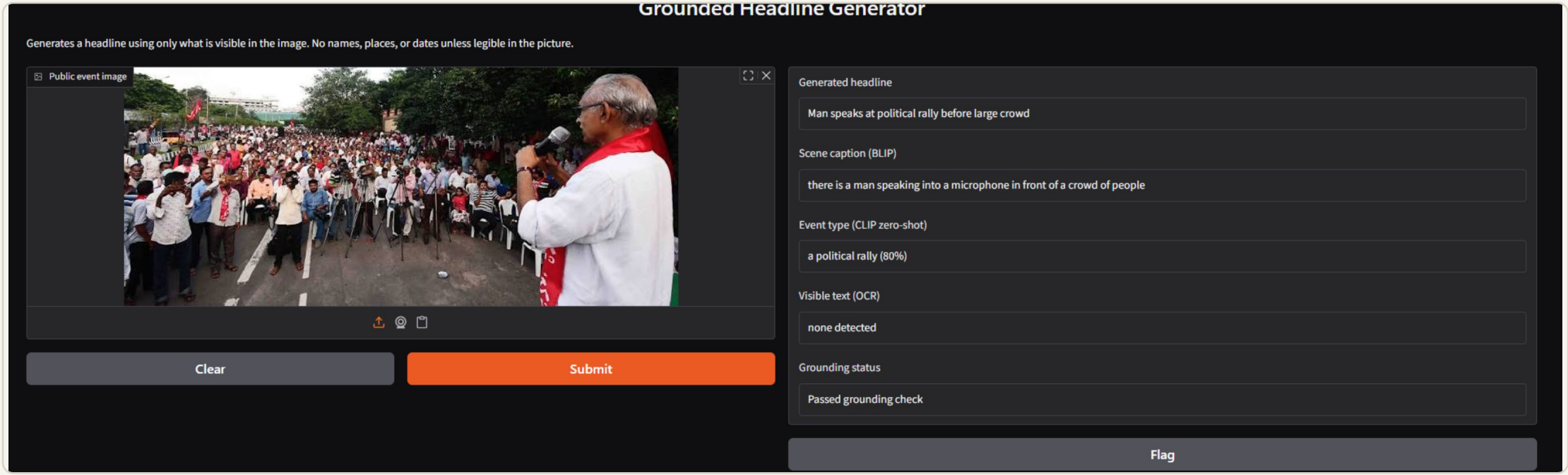
Pre-trained models, no training needed

Stage	Model or library	Source
Captioning	Salesforce/blip-image-captioning-large	Hugging Face
Event typing	openai/clip-vit-base-patch32	Hugging Face
Headline writer	Qwen/Qwen2.5-1.5B-Instruct	Hugging Face
Visible text	EasyOCR + pyspellchecker	GitHub / PyPI
Demo app	Gradio	gradio.app

Python, PyTorch and Transformers, run on Google Colab with a T4 GPU. Links to every model are in the README.

DEMO 1 · CLEAR SCENE

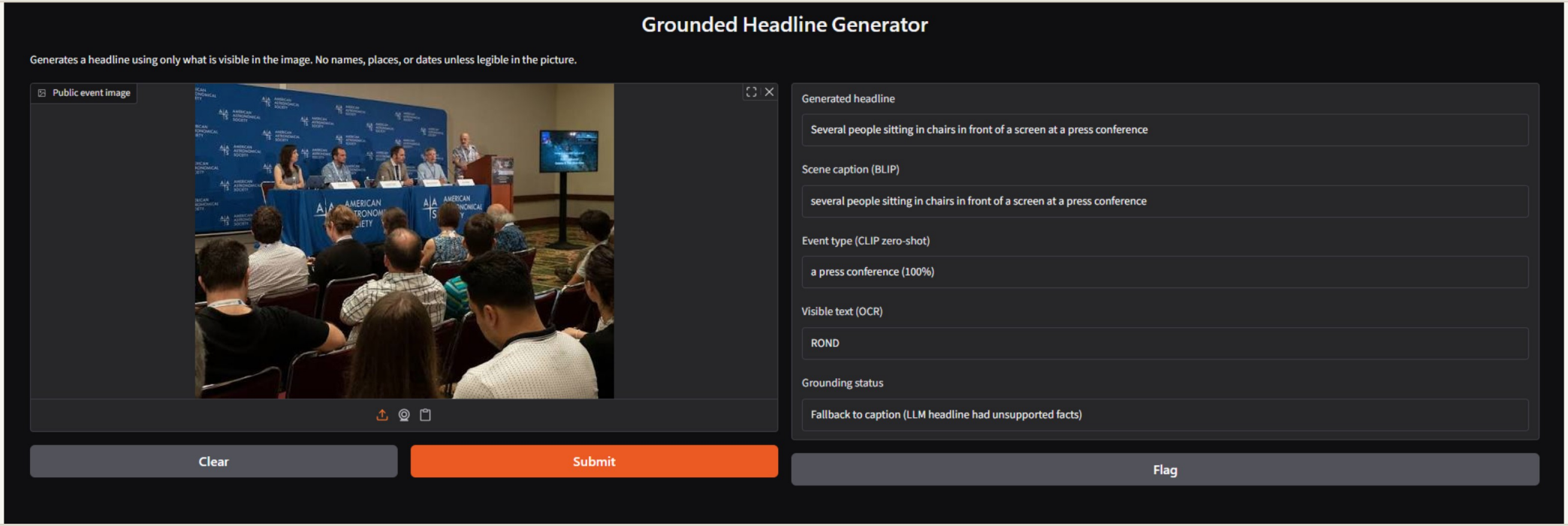
The headline passes the grounding check



Caption and event type are right (political rally, 80%). Headline: “Man speaks at political rally before large crowd”.

DEMO 2 · WEAK EVIDENCE

When the evidence is weak, it falls back safely



In an early version, OCR returned only noise (“ROND”) and the first headline contained unsupported content. The check rejected it and the system used the plain caption instead.

RESULTS ON 8 TEST IMAGES

Good on clear scenes, weaker on details

Scene	Generated headline	Our verdict
Political rally	Man speaks at political rally before large crowd	Good
Concert	Crowd watches band perform at concert	Good
Press conference	People sit in chairs during a press conference	Good
Market	A bustling market thrives with shoppers	Adjectives go beyond the image
Festival crowd	Aerial view shows religious festival crowd procession	“Religious” is a guess
Running race	Group of men run marathon race city street	“Marathon” is a guess
Sport	Men squatting during squash game	Doubtful
Climate march	Protesters march for fossil march	Wrong

No fallback in the final run. Manual review of 8 images; no accuracy score was computed.

LIMITATIONS

What the grounding check cannot catch

- **It compares words, not meaning.** On the climate march, banner fragments were merged into “Protesters march for fossil march”, and every word was in the evidence.
- Unsupported adjectives and causes, such as “bustling” and “thrives”, pass the check.
- OCR still misses large, stylised or angled banner text.
- The captioner can mislabel details such as the sport being played.
- The test set is small, so results are illustrative only.

THE TEAM

Who did what

Aditya Kumar

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Image captioning (BLIP) and event-type detection (CLIP). Collected the test images and ran the batch tests on Colab.

Aman Kumar

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Headline generation with Qwen2.5, the OCR stage and its noise filter, and the grounding check with retry and fallback.

Aman Kumar

2330361

Gradio demo app, screenshots, GitHub repository, report, slides and the presentation video.

CONCLUSION AND FUTURE SCOPE

A careful writer that knows when to stay general

What we built

- A working prototype with a live demo app
- Headlines built only from image evidence
- A check that rejects unsupported names and numbers

What comes next

- A single vision-language model such as Qwen2-VL
- A meaning-aware (NLI) faithfulness check
- Better banner OCR, Hindi support, larger evaluation

Code, report and sample outputs: github.com/Aditya-kr24/Headline_Generator